SPE-228918-MS

Appraising Fracture-Related Mud Losses During Drilling: Fact or Fiction

A. Bin Ab Fatah, L. Malla, and M. Al Shemsi, ADNOC Onshore

 DOI 10.2118/228918-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Mud losses during drilling have significantly increased the cost of wells, particularly in extended-length horizontal wells. It is widely understood that natural fractures and faults contribute to these mud losses. This solution leverages on Natural Language Processing (NLP) technique to extract structured information from unstructured information in the drilling reports seeks to understand the causes and implications of these losses and to propose solutions for mitigating such issues in future drilling operations.

The study interprets seismic data to identify faults and fractures, using Gradient Coherency Cube and Variance attributes to highlight features and confirm seismic interpretations with borehole image data, Production Logging Tool (PLT), production and injection data, Extended Leak Off (XLOT) and Micro-Frac tests and pressure measurements.

Analyzing mud losses around the offset wells reveals their extent and severity. Natural Language Processing (NLP), the process of automatically extracting structured information from unstructured text is used to simplify and expedite the mundane task of extracting information from drilling reports. This involves identifying and pulling out specific pieces of data like names, dates, locations, and relationships, transforming vast amounts of text into organized, usable information from all the wells in the reservoir.

The analysis highlights the necessity of considering event sequences when linking mud losses to faults or fractures. It mentions that mud circulation might penetrate these features in earlier uncased sections, though most identified fractures are only partially conductive. A mud contamination issue starting at earlier depth caused pressure fluctuations and increased losses.

The presence of CO_2 and oil suggests that formation fluids have infiltrated the mud, altered its properties and reduced the mud weight. This, combined with pressure fluctuations and mechanical erosion, can degrade the mud filter cake, exposing the borehole to further contamination and losses. The significant increase in dynamic mud losses, while static loss remains low, indicates that mud may have opened natural and new fractures during circulation. Raising mud weight could create excessive overbalance, risking integrity of the formation.

Introduction

During the drilling of a horizontal water injector well in the Kharaib formation, the operation encountered multiple challenges like severe mud losses, ongoing contamination from CO_2 and crude oil, and significant pressure fluctuations throughout the drilling process. These complications adversely affected mud properties, compromised tool reliability, and hindered overall well delivery.

Mud losses in drilling, also known as lost circulation, occur when drilling fluid escapes into the formation instead of returning to the surface. There are several conditions and subsurface environments that can give rise to mud losses. For instance, formations characterized by high permeability, such as unconsolidated sands or loosely packed gravel beds, can readily absorb large volumes of drilling fluid. Similarly, naturally fractured formations where cracks, joints, or faults create interconnected pathways serve as conduits through which the mud can rapidly escape. Carbonate rocks, like limestone and dolomite, often contain cavernous or vuggy sections: these large voids, sometimes formed by the dissolution of minerals, can become significant loss zones. Additionally, formations that have experienced extensive previous production may be depleted, resulting in lower formation pressures. In such cases, the pressure differential between the wellbore and the formation can drive mud into these depleted zones more easily.

Excessive downhole pressures, whether due to overbalanced drilling practices or miscalculations in mud weight, can also induce fractures in otherwise competent formations. These newly created fractures provide fresh escape routes for the drilling fluid, compounding the problem of lost circulation. The operational consequences of mud losses are multifaceted. Not only do they lead to increased drilling costs through the loss of expensive drilling fluids and the need for additional materials and time to address the issue, but they can also jeopardize well control by making it harder to maintain the necessary hydrostatic pressure. Furthermore, excessive mud losses can have environmental ramifications, such as contamination of groundwater resources or surface spills.

To mitigate and control lost circulation, the industry employs a variety of specialized materials and engineering techniques. Lost circulation materials (LCMs), which can range from fibrous, granular, or flake materials to rapidly setting cements, are introduced into the drilling fluid to plug the loss zones and restore circulation. In severe cases, more advanced interventions such as placement of cement plugs or even the use of expandable casing may be necessary to seal off problematic formations and maintain wellbore integrity.

Natural fractures are frequently cited as the primary cause of mud losses; however, this is not always accurate, and thorough investigation is necessary to determine the actual causes. Gaining a comprehensive understanding of the factors and mechanisms behind mud losses whether they stem from natural geological formations, induced fractures, or pressure anomalies as it is crucial for developing effective drilling strategies and promoting both operational efficiency and environmental responsibility. This study seeks to achieve these objectives using Natural Language Processing techniques for information extraction.

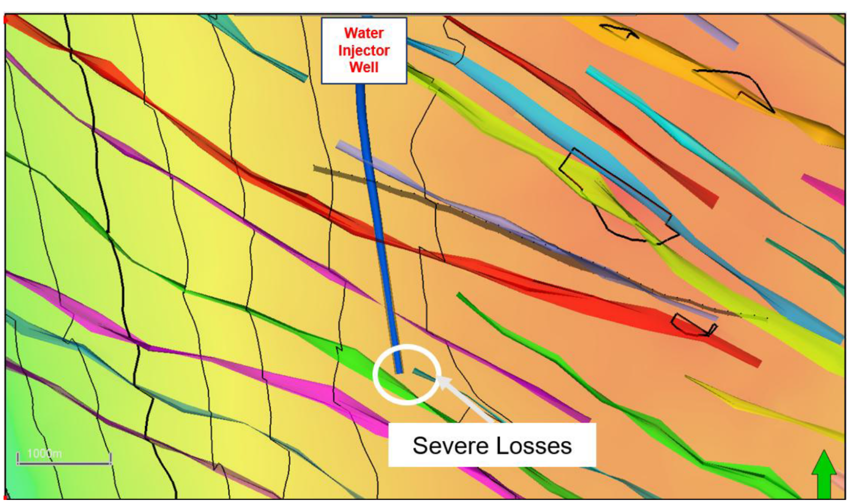

Figure 1—The horizontal water injector well is drilled in N-S direction and severe losses happened about 1000 ft before TD. The colorful lines are the faults interpreted based on seismic.

Background Geology

The Kharaib Formation in the western onshore Abu Dhabi represents a crucial component of the Lower Cretaceous carbonate system that dominates the subsurface geology of the Arabian Plate. Deposited during the Barremian to early Aptian stages, the Kharaib Formation is part of the Thamama Group and is bounded below by the Lekhwair Formation and above by the Shuaiba Formation. Its significance lies not only in its stratigraphic position but also in its role as a prolific hydrocarbon reservoir. The formation reflects a complex interplay of depositional environments, diagenetic processes, and sequence stratigraphy, all of which contribute to its reservoir heterogeneity and production potential.

The depositional setting of the Kharaib Formation was shaped by a broad, shallow carbonate platform that experienced varying degrees of restriction and open marine influence. This platform was subject to episodic sea-level changes, which controlled the stacking patterns of sedimentary units and the development of reservoir and non-reservoir facies. The formation comprises a range of lithofacies deposited in environments that include open marine ramps, restricted lagoons, and tidal flats. High-energy subtidal zones favored the accumulation of skeletal-peloidal packstones, algal-coated grainstones, and rudist floatstones, which now form the primary reservoir facies. In contrast, low-energy settings led to the deposition of bioturbated wackestones, organic-rich mudstones, and dense limestones, which act as baffles or barriers to fluid flow.

Stratigraphically, the Kharaib Formation is typically divided into three main units: the Lower Kharaib Reservoir Unit, the Middle Dense Zone, and the Upper Kharaib Reservoir Unit. These are capped by the Upper Dense Zone, often referred to as the Hawar Member. The reservoir units are characterized by relatively high porosity and permeability, while the dense zones are more compact and less permeable, serving as internal flow barriers. This stratigraphic architecture is critical for reservoir modeling and field development, as it influences the distribution of hydrocarbons and the efficiency of recovery strategies. The presence of stylolites, firmgrounds, and other diagenetic features further complicates the reservoir framework, necessitating detailed geological and petrophysical analysis (Strohmenger et al., 2004)

Diagenesis has played a pivotal role in shaping the reservoir quality of the Kharaib Formation. Early marine cementation, compaction, and burial diagenesis have altered the original depositional textures, enhancing or reducing porosity depending on the facies and burial history. Stylolites, often associated with pressure solutions, mark sequence boundaries and can act as barriers to vertical fluid flow.

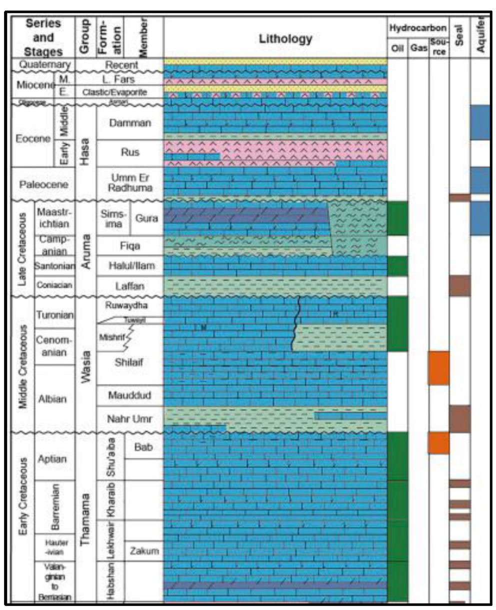

Figure 2—Composite stratigraphic column of UAE shows Kharaib as one of the major carbonate reservoirs (Yin et al., 2018).

Natural Fractures

The fracture systems in Kharaib are influenced by both regional tectonics and local stress regimes, resulting in distinct orientations and intensities across the formation. Studies have shown that fracture intensity and density vary vertically within the Kharaib Formation, largely controlled by contrasts in rock texture. Dense carbonate layers, often associated with stylolitic horizons or compacted mud-rich intervals, tend to be more heavily fractured than the more porous grain-supported facies. This vertical heterogeneity is critical for understanding fluid flow, as fractures in dense zones can act as conduits for hydrocarbons, connecting isolated porous bodies and enhancing overall reservoir connectivity. In the field, two dominant fracture sets have been observed: one trending NW-SE and another NNE-SSW, which aligns with the present-day in-situ stress field. These orientations are consistent with regional tectonic trends and have been confirmed through borehole image logs and well test data, which show preferential flow along these fracture directions at the graben area.

Fractures in the Kharaib Formation are not uniformly distributed; instead, they tend to cluster around structural features such as faults, stylolite boundaries, and firmgrounds. These clusters often coincide with zones of enhanced porosity and permeability. Overall, the natural fracture network within the Kharaib Formation adds a critical layer of complexity to reservoir characterization and modeling. Understanding the geometry, orientation, and connectivity of these fractures is essential for accurate simulation and effective field development.

Seismic Interpretation

Seismic data plays a crucial role in subsurface imaging, aiding in the identification of geological structures and features. A key technique employed in this study involved the use of an advanced seismic interpretation tool. This tool enhances the understanding of subsurface geological features through advanced seismic attributes such as Gradient Coherence Cube (GCC) blended with conventional seismic Variance attributes. These techniques provide clearer and more detailed images of geological features than traditional seismic attributes.

The GCC is utilized in geophysical exploration to enhance subsurface feature visualization. Derived from the coherence cube, which measures similarity between seismic traces to detect discontinuities such as faults and stratigraphic features, the GCC focuses on changes in coherence values, highlighting subtle variations in the seismic data (Massala et al., 2013). This attribute assists geophysicists in identifying and interpreting complex geological structures, including fractures, channels, and other stratigraphic anomalies.

These advanced seismic attributes were complemented by Relative Geological Time (RGT) slicing, enabling efficient and effective screening of seismic attribute volumes without manual interpretation. RGT facilitates auto-tracking of all possible horizons within a seismic volume, allowing for the automatic generation of a continuous model of geological layers based on chronostratigraphy (Lacaze et al., 2016). Stratal slicing performed through RGT enabled the analysis of geological features at different time intervals. The combination of RGT with GCC technique allowed for the extraction of fine geomorphological features from seismic data that might be missed using classical methods. This provided high-resolution seismic images crucial for identifying minor faults.

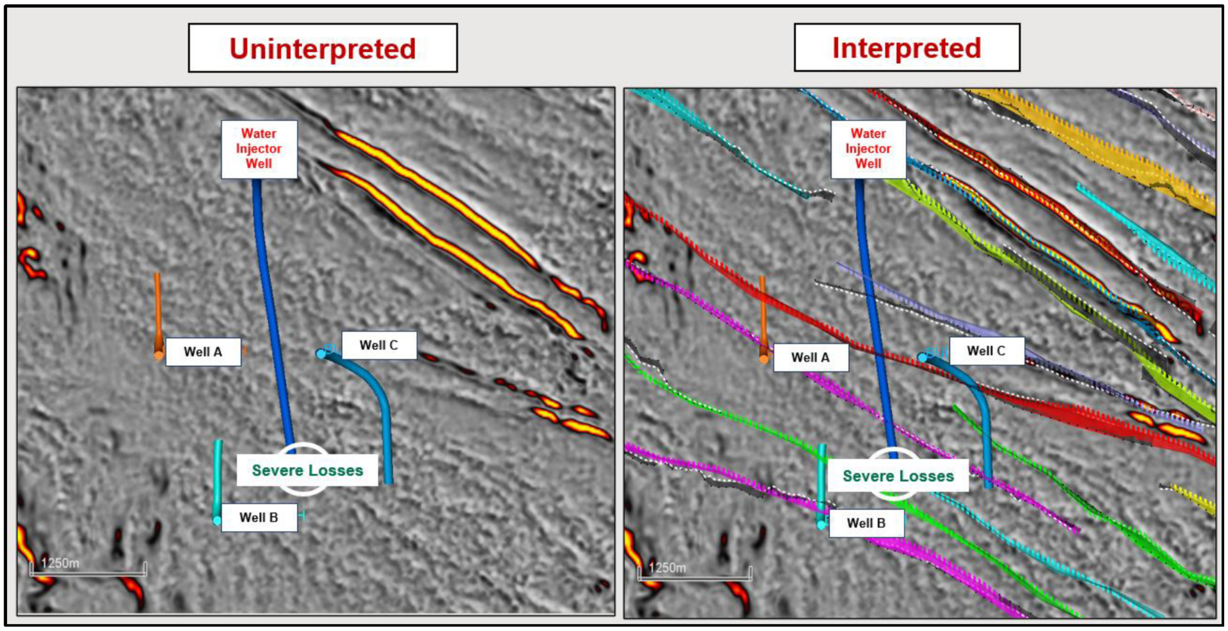

Figure 3—The figure on the left shows the uninterpreted map on GCC blended with Variance attributes. It highlights the faults and fractures. The figure on the right shows the interpreted faults on the map.

Mud Losses Data Extraction

This study utilizes a rigorous and systematic methodology, employing Natural Language Processing (NLP) to extract unstructured data from drilling reports for the analysis of mud loss incidents in drilling operations. The first step involved the collection of over 150 PDF-based well reports from the field. These reports, which documented various drilling campaigns with recorded mud loss incidents, provided a rich dataset for analysis.

To extract the raw unstructured text from these reports, PyMuPDF was utilized. This tool was chosen for its efficiency in handling large volumes of PDF documents and its ability to retain spatial references such as well, depth, and date. The extracted text served as the foundation for the subsequent data processing steps. The choice of PyMuPDF was driven by its robust performance in preserving the context of the extracted data, which is crucial for accurate analysis.

Named Entity Recognition (NER) was then performed using spaCy with a custom NER model. This model was fine-tuned on labeled drilling terminology to accurately extract relevant entities such as mud loss volume, depth, formation name, mud weight, and pressure events. The use of a transformer-based NER model ensured high precision and recall in identifying these entities from the unstructured text. The customization of the NER model involved training it on a dataset specifically annotated with drilling-related terms, which significantly improved its performance in this domain.

Once the entities were extracted, the data was structured and cleaned using Pandas and regular expressions. This involved standardizing numerical units (e.g., barrels, psi, feet) and filtering out duplicate and null entries. This step was crucial to ensure the consistency and reliability of the dataset for subsequent analysis. The data cleaning process also included the removal of any inconsistencies and errors that could potentially skew the results. The structured data was then organized into a tabular format, making it easier to analyze and visualize.

Visualization of the structured data was achieved using Plotly, Matplotlib, and GeoPandas. Bubble maps were created to represent the volume of mud loss (bubble size) and the severity of the loss (bubble color). Geospatial coordinates were extracted from internal well trajectory datasets to accurately plot the locations of mud loss incidents on the map. The bubble maps provided a clear and intuitive way to visualize the spatial distribution of mud losses, highlighting areas with the highest severity. The use of interactive visualization tools like Plotly allowed for dynamic exploration of data, enabling users to zoom in on specific regions and examine the details of individual mud loss events.

To identify fault corridors, GCC and Variance attributes were employed. These seismic attributes were selected for their effectiveness in highlighting subsurface fault structures. The interpreted fault zones were then overlaid on the mud loss maps to visualize the spatial correlation between mud losses and fault corridors. The integration of seismic data with mud loss information provided valuable insights into the underlying causes of mud losses and helped identify high-risk areas. The predominant trend in bubble map indicates that significant mud loss is concentrated in the central region, particularly within the graben area, while lower losses are observed at the flanks. Although some substantial losses are associated with faults and fractures, this correlation does not consistently apply.

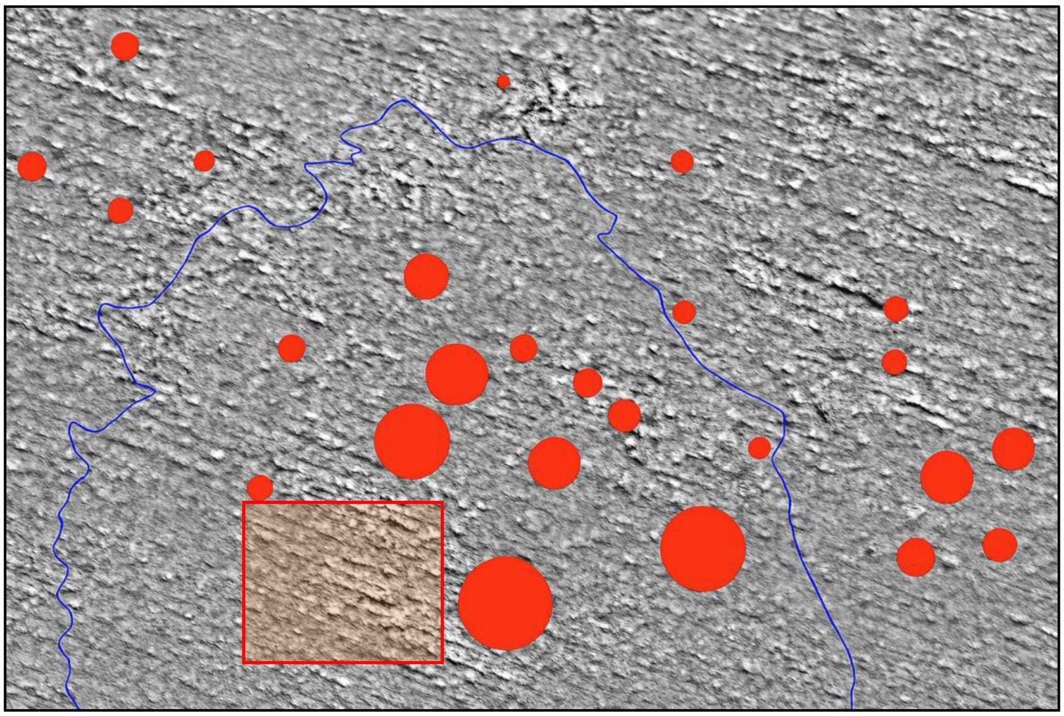

Figure 4—The bubble map shows the severity of the mud losses based on bubbe size and overlaid on GCC map to check the position relatives to the faults and fractures. The red circles are losses which are more than 100 bbl/hr and the red box is the vicinity of the drilled well.

Integration with other Data

The horizontal water injector well, strategically situated in the central-western sector of the field, follows a north-to-south trajectory that intersects four significant fault lines. Throughout its course, the well encounters a diverse range of subsurface features, with losses observed at a location approximately ~1000 feet short of the planned total depth (TD). This spatial relationship between the wellbore and the intersected faults is crucial, as such intersections often heighten the risk of drilling complications, particularly mud loss incidents.

Faults identified during seismic and mud losses map analysis were categorized based on the confidence level derived from combined seismic and borehole data. This graded approach to fault classification aids in prioritizing zones for further investigation and risk mitigation. To enrich the dataset, Production Logging Tool (PLT) measurements were integrated into the evaluation, providing valuable insights into the types and movements of fluids along the wellbore. These measurements are instrumental in diagnosing production anomalies and understanding the vertical and lateral distribution of fluid entries and exits within the reservoir. The PLT data, covering multiple horizontal injector and producer wells, revealed that apart from wells intersecting prominent faults in the graben (depressed) areas, there was no systematic correlation between fluid movement and fault presence.

The BHI analysis further substantiated the existence of two faults initially interpreted from seismic data, establishing a strong link between geophysical prediction and direct borehole observation. The identified conductive and partially conductive fractures within the BHI logs were primarily found in sections where the well's inclination changed abruptly. This suggests a close relationship between drilling mechanics and fracture development, with many of these fractures likely attributed to the stresses induced during drilling rather than pre-existing natural fractures. Notably, mud contamination, an early indication of wellbore instability and fluid losses, as first detected at earlier depth, continuing into the portion of the well where losses became more severe.

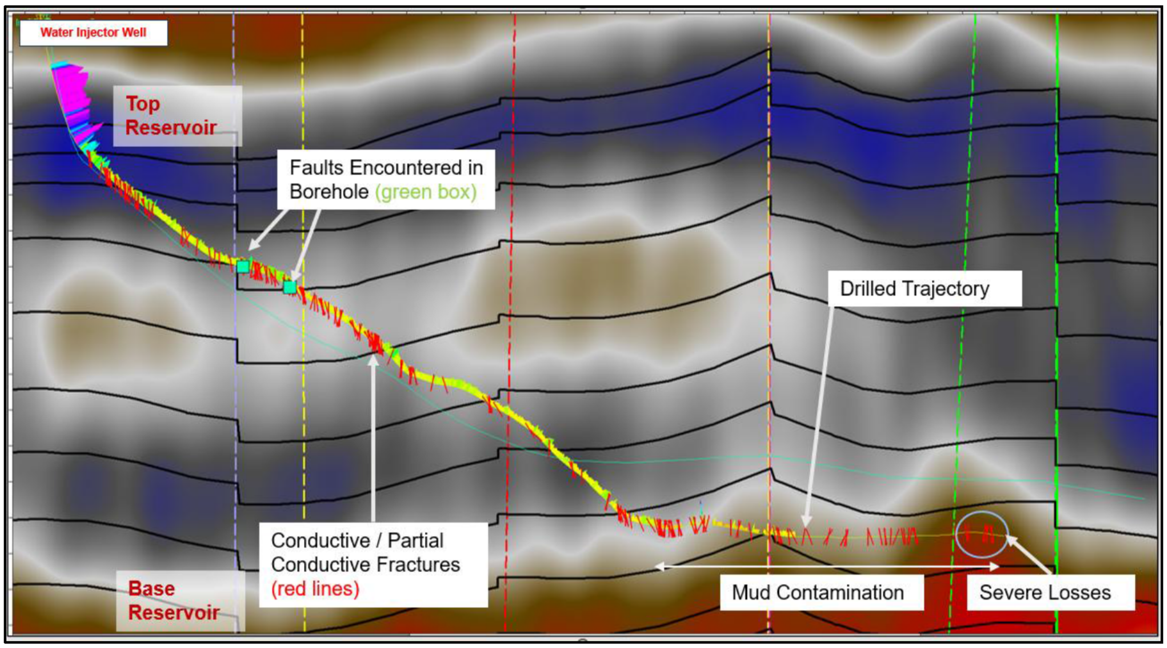

Figure 5—The well trajectory with GR log is overlaid on the seismic section with fractures and faults on the well. Mud contamination and severe losses areas are highlighted to show the sequence of event.

In the upper portion of the analyzed wellbore, a principal fault was observed, accompanied by several smaller faults, all of which were associated with conductive fractures. Conversely, the lower section contained only partially conductive fractures, indicating a possible change in fracture openness or connectivity with depth. Both seismic and BHI interpretations were consistent in identifying the fault strike orientation as northwest-southeast (NW-SE), providing a coherent geological framework for fault distribution. Geosteering data, collected in real time during drilling, further corroborated the existence of a seismic-interpreted fault, measuring a 14-foot vertical displacement (throw), though this specific feature was not linked to significant fluid losses.

A comprehensive review of eleven horizontal wells within a two-kilometer radius of the water injector well provided context for understanding local reservoir behavior. These wells, comprising producers and injectors offered additional data points for comparison. Some nearby oil producers exhibited associated gas production, and minor CO_2 contamination was observed, likely originating from these adjacent wells. In one instance, the drilled well was contaminated with oil, supporting the hypothesis of reservoir connectivity and potential crossflow between wells. Analysis of the reservoir pressure gradient using offset wells indicated a progressive increase from the outer flanks toward the central field, exceeding 1000 psi between the nearby offset wells. Such gradients are significant, as they contribute to the dynamic conditions that can precipitate mud losses during drilling.

Geomechanical analysis, informed by a 1D model and supported by Extended Leak Off Test (XLOT) and Micro-Fracture (Micro-Frac) testing, provided critical insight into formation strength and stress regimes. The applied mud weight during drilling was 11.2 pounds per gallon (ppg), equating to a pressure gradient of 0.58 psi/ft under static conditions. However, when accounting for the Equivalent Circulating Density (ECD) which includes annulus pressure losses due to fluid movement, the effective pressure exceeded the average minimum horizontal stress (SH_{min}), measured at 0.57 psi/ft. SH_{min} is often used interchangeably with Fracture Closure Pressure (FCP), representing the threshold at which fractures in the formation will close under pressure. Surpassing this threshold, especially during dynamic conditions such as fluid circulation, increases the likelihood of inducing new fractures within the formation, thereby contributing to the observed mud losses.

This integrated, multidisciplinary approach of combining advanced seismic interpretation, borehole imaging, production logging, and geomechanical analysis provides a comprehensive understanding of the factors influencing mud loss during drilling. The findings emphasize the complexity of interactions between geological structures, reservoir properties, and operational parameters, underscoring the importance of data-driven decision-making in mitigating drilling risks and improving well performance.

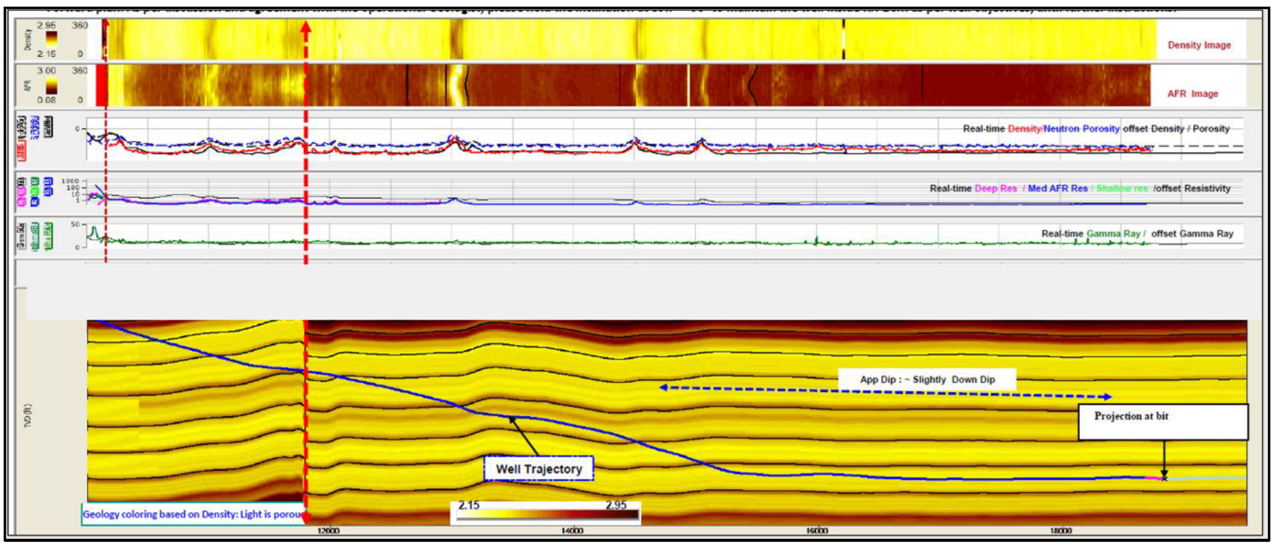

Figure 6—Geosteering data confirmed one of the faults interpreted in seismic in upper section. No losses happened even though the throw seen in geosteering was 14 ft.

Conclusion

Using NLP to extract unstructured data from drilling reports streamlines learning from well incidents. The bubble map interface provides an intuitive way for subsurface team to visualize subsurface risk during prognosis. This integrated approach helps Geoscientists, Petroleum Engineers, and Drilling Engineers prognose the possible mud loss, detect early signs of wellbore instability and quickly adjust drilling parameters and mud weights, ensuring smooth operations and managing uncertainties or deviations from fault and fracture forecasts.

The integrated analysis confirms the presence of one major fault and one minor fault, which were previously identified in the seismic data. Conductive and partially conductive fractures were observed throughout the section, particularly in regions with rapid changes in well inclination. This suggests that many of these fractures may have been created during the drilling process. To effectively link the losses to faults or fractures, it is crucial that these losses also occur earlier in the section rather than only towards the end, to prevent biased interpretations. However, this is not the case even though a 14-foot fault throw is observed in the upper section. There is a chance that mud circulation could penetrate these faults and fractures in the earlier uncased section; however, most of the identified conductive fractures are partially conductive as opposed to fully conductive.

Additionally, an issue with mud contamination was noted starting at earlier depth, as highlighted by the driller's remarks. The driller observed excessive pressure fluctuations due to the mud contamination and increased the mud weight from 11 to 11.2 ppg. Despite this, the presence of CO_2 and oil traces persisted, causing pressure fluctuations and increasing the losses. At further depth, the contamination became severe. The detection of CO_2 and oil contamination suggests that formation fluids have infiltrated the mud, changed its rheological properties and caused it to become less viscous. This led to the need for increased mud weight. Under these conditions, pressure fluctuations and mechanical erosion from rapid drill string movement can result in the degradation of the mud filter cake, thereby exposing the borehole to reservoir formation fluids, leading to contamination and losses.

Furthermore, raising the mud weight may create excessive overbalance, jeopardizing the integrity of the formation. The conclusions emphasize the need to avoid biased interpretations as the nearby wells history was not properly reviewed leading to underestimation of contamination risk and lack of appropriate fluid formulation & monitoring systems. Overall, the analysis highlights the importance of considering the sequence of events in linking losses to faults and fractures, as well as the potential role of mud circulation in penetrating these geological features.

References

Lacaze, S., Adam, J.P., Durot, B., Pauget, F., Global Technique in Seismic Interpretation for Reservoir Detection and Characterization, 2016, 12th Middle East Geoscience Conference & Exhibition, Manama, Bahrain, 7-10 March 2016.

Massala, A., Keskes, N., Goncalves-Ferreira, L., Onu, C., Cordier, P., 2013, An Innovative Attribute for Enhancing Faults Lineaments And Sedimentary Features During 2G&R Interpretation, SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, 30 September–2 October 2013.

Strohmenger, C.J., Ghosh, D., Weber, L.J., Ghani, A., Rebelle, M., Al-Mehsin, K., Al-Jeelani, O., Al-Mansoori, A., Suwaina, O., 2017, High-Resolution Sequence Stratigraphy of the Kharaib Formation (Lower Cretaceous, U.A.E.), Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 10-13 October 2004.

Yin, S.Z., Bai G.P., Gao, L., Xu, C.Y., A facies and palaeogeography-based approach for analysis of petroleum systems in United Arab Emirates, *Journal of Palaeogeography*, 2018, **7**(2): 168–178 (00145)